

Attained Three-Point Moment Selection for Five Labels with Direction Multiplicities $3 + 1 + 1$

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Abstract

We study a finite complex moment system on five labelled atoms and prove a restricted three-point selection result. If the unit directions take exactly three distinct values with multiplicities $3, 1, 1$, then some probability vector supported on at most three labels is admissible and attains selection cost at most $5/8$ of the original response energy. The proof combines an exact reduction to distances from two targets to the edges of a weighted triangle with a sharp enough edge-distance inequality. This is a local result: the five-label problem for four or five distinct directions, the six-label four-point problem, and the full seven-atom conjecture all remain unresolved.

1 Introduction and scope

Zhang’s circle-moment formulation of weighted data selection reduces the smallest unresolved vector-valued regression cell to two inequalities for finite moment systems [Zha26, Section 7]. In that formulation, Conjecture 7.8 asks for infimum bounds with support budgets three and four, and Proposition 7.9 reduces the general question to at most seven atoms. The same source proves the three-point inequality for systems with at most four atoms (with the stronger bound $E/2$) and the four-point inequality for systems with at most five atoms (with bound $2E/9$); see [Zha26, Section 7.6.5].

Here we isolate and solve one natural five-label stratum of the three-point problem. Besides increasing the label count in this stratum, we prove literal attainment by an admissible probability vector, rather than only an infimum statement. The restriction appears in the title, abstract, and theorem: exactly three directions occur, and one direction contains three labels while each other direction contains one. No conclusion is asserted for another direction pattern.

There is a broad methodological parallel with the unrelated order-five correlation-matrix problem of Zeng, Liu, and Ratnavelu [ZLR26]: that work also reduces a finite rank-two problem to unimodular variables and closes the remaining inequalities with exact algebraic certificates. It neither states nor implies any result about the moment-selection system considered here.

2 Moment systems and the local theorem

A *finite moment system* is a family

$$(p_i, t_i, z_i)_{i=1}^5, \quad p_i > 0, \quad \sum_{i=1}^5 p_i = 1, \quad t_i, z_i \in \mathbb{C}, \quad |t_i| = 1,$$

satisfying

$$\sum_i p_i t_i = 0, \quad \sum_i p_i z_i = 0, \quad \sum_i p_i \bar{t}_i z_i = 0. \quad (1)$$

Its response energy is

$$E = \sum_i p_i |z_i|^2.$$

For a probability vector $q = (q_i)_{i=1}^5$, put

$$\mu = \sum_i q_i t_i, \quad \nu = \sum_i q_i z_i, \quad \lambda = \sum_i q_i \bar{t}_i z_i.$$

We call q *admissible* when $|\mu| < 1$. In that case define $(a, b) \in \mathbb{C}^2$ by

$$\begin{pmatrix} 1 & \mu \\ \bar{\mu} & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \nu \\ \lambda \end{pmatrix}, \quad C(q) = |a|^2 + |b|^2. \quad (2)$$

Theorem 2.1 (The 3 + 1 + 1 stratum). *Suppose a five-label moment system satisfying (1) has exactly three distinct direction values, with multiplicities 3, 1, 1. Then there is an admissible probability vector q such that*

$$|\text{supp } q| \leq 3, \quad C(q) \leq \frac{5}{8}E.$$

In particular, the bound is attained by a permitted selection; it is not merely a bound on an infimum.

The proof occupies Sections 3–5.

3 Three-direction identities and attained candidates

Let G_1 be the direction class containing three labels, and let G_2, G_3 be the singleton classes. Write their distinct unit directions as τ_1, τ_2, τ_3 , and set

$$P_g = \sum_{i \in G_g} p_i, \quad m_g = P_g^{-1} \sum_{i \in G_g} p_i z_i.$$

Only G_1 has within-class variation. Put

$$r_i = \frac{p_i}{P_1}, \quad w_i = z_i - m_1 \quad (i \in G_1), \quad V = \sum_{i \in G_1} r_i |w_i|^2,$$

and

$$E_m = \sum_{g=1}^3 P_g |m_g|^2.$$

Thus $\sum_{i \in G_1} r_i w_i = 0$.

Lemma 3.1 (Class-energy identities). *For $g = 1, 2, 3$,*

$$E = E_m + P_1 V, \quad P_g |m_g|^2 = (1 - 2P_g) E_m. \quad (3)$$

Moreover $0 < P_g < 1/2$.

Proof. The vectors

$$u = (\sqrt{P_g})_{g=1}^3, \quad v = (\sqrt{P_g} \tau_g)_{g=1}^3$$

are orthonormal in $\mathbb{C}^3$, because $\sum_g P_g = \sum_g P_g |\tau_g|^2 = 1$ and $\sum_g P_g \tau_g = 0$. The last two equations in (1) say that

$$y = (\sqrt{P_g} m_g)_{g=1}^3$$

is orthogonal to both u and v . Their orthogonal complement is one-dimensional, and the diagonal entries of its projection $I - uu^* - vv^*$ are $1 - 2P_g$. If $y = 0$, the second identity in (3) is immediate. Otherwise, the squared coordinate proportions of y are $1 - 2P_g$, which proves that identity. The first identity is the within-between variance decomposition.

Finally,

$$P_g = \left| \sum_{k \neq g} P_k \tau_k \right| \leq 1 - P_g.$$

Equality would force the other two distinct directions to be parallel, which is impossible. Thus $P_g < 1/2$. $\square$

We next obtain two families of selections. Denote

$$d_*^2 = \min_{i \in G_1} |w_i|^2, \quad \Gamma = \bigcup_{\substack{i, j \in G_1 \\ i < j}} [w_i, w_j].$$

Proposition 3.2 (Exact attained candidates). *There is an admissible selection on three labels with cost*

$$A_0 = 2P_1^2 d_*^2. \quad (4)$$

For $g \in \{2, 3\}$, let h be the other singleton index and define

$$\delta_g = \text{dist} \left(-\frac{m_1}{2P_g}, \Gamma \right).$$

There is also an admissible selection on at most three labels attaining

$$S_g = \frac{1 - 2P_g}{2P_g} E_m + \frac{2P_1 P_g}{1 - 2P_h} \delta_g^2. \quad (5)$$

Consequently an admissible support- ≤ 3 selection attains

$$\min\{A_0, S_2, S_3\}.$$

Proof. For (4), choose a label $i \in G_1$ attaining d_* , retain the two singleton labels, and assign the three selected labels masses P_1, P_2, P_3 . Then $\mu = 0$, while the two right-hand sides of (2) are $P_1 w_i$ and $P_1 \bar{w}_i$. Hence the cost is exactly $2P_1^2 |w_i|^2$.

Fix $g \in \{2, 3\}$. Any probability vector giving positive total mass to the two distinct directions τ_1, τ_g is admissible. If its conditional response at τ_1 is $m_1 + w$, its coefficient vector (a, b) is the unique two-point interpolant

$$a + \tau_1 b = m_1 + w, \quad a + \tau_g b = m_g.$$

Let A_g denote the interpolant for $w = 0$. Changing m_1 by w changes the coefficient vector by

$$v_g w, \quad v_g = \frac{(-\tau_g, 1)}{\tau_1 - \tau_g}.$$

Eliminating m_h from the two aggregate response moments and using $\sum_k P_k \tau_k = 0$ gives

$$(1 + \overline{\tau_g} \tau_1) m_g = -\frac{1 - 2P_g}{P_g} m_1. \quad (6)$$

A direct orthogonal decomposition of $A_g + v_g w$, using (6), yields

$$\|A_g + v_g w\|^2 = \frac{|m_g|^2}{2} + \frac{2}{|\tau_1 - \tau_g|^2} \left| w + \frac{m_1}{2P_g} \right|^2. \quad (7)$$

For completeness, one may verify (7) directly from

$$A_g + v_g w = \frac{1}{\tau_1 - \tau_g} \begin{pmatrix} \tau_1 m_g - \tau_g (m_1 + w) \\ m_1 + w - m_g \end{pmatrix}$$

and (6).

Splitting the selected mass in direction τ_1 between any two labels realizes every w on the corresponding segment $[w_i, w_j]$. Taking, for example, total mass $1/2$ on that direction and mass $1/2$ on the singleton in G_g makes the resulting q admissible. Segment endpoints use only one label from G_1 . Compactness of the three segments therefore shows that the minimum in (7) is attained.

It remains to rewrite its value. Lemma 3.1 gives

$$|m_g|^2 = \frac{1 - 2P_g}{P_g} E_m.$$

Taking squared moduli in $P_1 \tau_1 + P_g \tau_g = -P_h \tau_h$ gives

$$P_1 P_g |\tau_1 - \tau_g|^2 = 1 - 2P_h. \quad (8)$$

Substitution into (7) proves (5). □

4 A weighted-triangle edge-distance bound

The geometric input below also covers collinear triples and targets outside the triangle.

Lemma 4.1 (Weighted triangle). *Let $r_i > 0$, $\sum_{i=1}^3 r_i = 1$, and $w_i \in \mathbb{C} \simeq \mathbb{R}^2$ satisfy*

$$\sum_{i=1}^3 r_i w_i = 0, \quad V = \sum_{i=1}^3 r_i |w_i|^2 > 0.$$

If $\Gamma = [w_1, w_2] \cup [w_1, w_3] \cup [w_2, w_3]$, then every $u \in \mathbb{C}$ satisfies

$$\text{dist}^2(u, \Gamma) \leq \begin{cases} \frac{(V + |u|^2)^2}{4V}, & |u|^2 \leq V, \\ |u|^2, & |u|^2 \geq V. \end{cases} \quad (9)$$

Proof. First suppose that the triangle is nondegenerate. The positive weighted barycentre condition puts 0 in its interior. For the side opposite w_i , let ℓ_i be its length, n_i its inward unit normal, $h_i > 0$ the distance from 0 to its supporting line, and A the triangle's area. Barycentric coordinates and the closed-polygon identity give

$$r_i = \frac{\ell_i h_i}{2A}, \quad \sum_i \ell_i n_i = 0. \quad (10)$$

Set

$$Z = \sum_i \frac{r_i}{h_i^2}, \quad \alpha_i = \frac{r_i/h_i^2}{Z}, \quad H = \frac{1}{Z}, \quad M = \sum_i \alpha_i n_i n_i^\top.$$

Then $\sum_i \alpha_i = 1$, $\text{tr } M = 1$, and (10) implies

$$\sum_i \alpha_i h_i n_i = 0, \quad \sum_i \alpha_i h_i^2 = H. \quad (11)$$

We claim the exact identity

$$H = V \det M. \quad (12)$$

Indeed, the weighted variance identity and (10) yield

$$V = r_2 r_3 \ell_1^2 + r_1 r_3 \ell_2^2 + r_1 r_2 \ell_3^2 = 4A^2 Z r_1 r_2 r_3,$$

where ℓ_i is the side opposite w_i . The 2×3 Cauchy–Binet formula, together with $|\det(n_i, n_j)| = 2A/(\ell_i \ell_j)$, gives

$$\det M = \sum_{i < j} \alpha_i \alpha_j \det(n_i, n_j)^2 = \frac{1}{4A^2 Z^2} \sum_{i < j} \frac{1}{r_i r_j} = \frac{1}{4A^2 Z^2 r_1 r_2 r_3} = \frac{1}{ZV}.$$

This proves (12).

If u lies in the triangle, its distance to the boundary equals the least of its three nonnegative signed side distances $h_i + n_i \cdot u$. Indeed, for a minimizing side the perpendicular foot remains in every defining half-plane, since all other signed distances are at least the distance moved. Thus (11)–(12) imply

$$\text{dist}^2(u, \Gamma) \leq \sum_i \alpha_i (h_i + n_i \cdot u)^2 = V \det M + u^\top M u. \quad (13)$$

Write the eigenvalues of the positive semidefinite trace-one matrix M as $\lambda, 1 - \lambda$, where $0 \leq \lambda \leq 1/2$, and put $U = |u|^2$. The right-hand side of (13) is at most

$$(1 - \lambda)(V\lambda + U).$$

Its maximum for $0 \leq \lambda \leq 1/2$ is $(V + U)^2/(4V)$ when $U \leq V$, and U when $U \geq V$.

If u is outside the triangle, the nearest point of the triangle lies on its boundary; since 0 lies in the triangle, $\text{dist}(u, \Gamma) \leq |u|$. For $U \leq V$, the first expression in (9) is at least U , so this estimate also gives the claimed branch. Finally, if the three points are collinear, their positive weighted barycentre implies that one of the three segments contains 0. Again $\text{dist}(u, \Gamma) \leq |u|$, completing all cases. $\square$

5 Proof of the local theorem

Proof of Theorem 2.1. If $E = 0$, positivity of the p_i forces every $z_i = 0$. Any selection using two distinct directions with positive weights is admissible and has zero cost. Hence assume $E > 0$.

Put

$$x_k = 1 - 2P_k \quad (k = 1, 2, 3).$$

By Lemma 3.1,

$$x_k > 0, \quad x_1 + x_2 + x_3 = 1, \quad E = E_m + P_1 V, \quad P_1 |m_1|^2 = x_1 E_m.$$

Proposition 3.2 supplies the attained value $A_0 = 2P_1^2 d_*^2$. If $A_0 \leq 5E/8$, we are done. Suppose instead that

$$2P_1^2 d_*^2 > \frac{5}{8}E. \quad (14)$$

Since $d_*^2 \leq V$, this forces

$$2P_1^2 V > \frac{5}{8}(E_m + P_1 V). \quad (15)$$

In particular $V > 0$, $P_1 > 5/16$, and, on writing

$$x = x_1, \quad y = \max\{x_2, x_3\}, \quad \zeta = \min\{x_2, x_3\},$$

we have

$$0 < x < \frac{3}{8}, \quad y \geq \frac{1-x}{2}, \quad \zeta = 1 - x - y. \quad (16)$$

In (5), choose h so that $x_h = y$, and choose g so that $x_g = \zeta$.

First suppose $E_m = 0$. Then $m_1 = 0$, so the target defining δ_g is 0. Lemma 4.1 gives $\delta_g^2 \leq V/4$. Formula (5) therefore gives

$$\frac{S_g}{P_1 V} \leq \frac{1 - \zeta}{4y} = \frac{x + y}{4y} < \frac{5}{8}.$$

The last inequality follows from $2x < 3y$, which is a consequence of (16).

It remains to consider $E_m > 0$. By homogeneity in the response values, normalize $E_m = 1$. Inequality (15) is then equivalent to

$$V > V_0 := \frac{10}{(1-x)(3-8x)}. \quad (17)$$

For the chosen singleton g , the squared norm of the target in Proposition 3.2 is

$$U := \left| \frac{m_1}{2P_g} \right|^2 = \frac{2x}{(1-x)(1-\zeta)^2}. \quad (18)$$

Since $1 - \zeta = x + y \geq (1+x)/2$, equations (17)–(18) imply

$$\frac{U}{V} < \frac{U}{V_0} \leq \frac{4x(3-8x)}{5(1+x)^2} < \frac{1}{2}. \quad (19)$$

The final inequality is equivalent to $5 - 14x + 69x^2 > 0$, whose discriminant is negative.

The first branch of Lemma 4.1 and (19) now give

$$\delta_g^2 \leq \frac{(V+U)^2}{4V} = \frac{V}{4} + \frac{U}{2} + \frac{U^2}{4V} \leq \frac{V}{4} + \frac{5}{8}U. \quad (20)$$

Using $P_k = (1 - x_k)/2$, substitute (18) and (20) into (5):

$$S_g \leq \frac{\zeta}{1 - \zeta} + \frac{5x}{8y(1 - \zeta)} + \frac{(1 - x)(1 - \zeta)}{8y} V. \quad (21)$$

The coefficient of V in the difference between

$$\frac{5}{8}E = \frac{5}{8} \left(1 + \frac{1 - x}{2} V \right)$$

and the right-hand side of (21) is

$$\frac{(1 - x)(3y - 2x)}{16y} > 0. \quad (22)$$

It therefore suffices to evaluate that difference at $V = V_0$. After multiplication by the positive denominator $8y(x + y)(3 - 8x)$, its numerator is

$$N = 30x^2 - 15x - 24y + 108xy + 54y^2 - 104xy(x + y). \quad (23)$$

Write $y = (1 - x)/2 + \rho$, with $\rho \geq 0$. Then

$$N = \left(\frac{3}{2} - 2x - \frac{21}{2}x^2 + 26x^3 \right) + (30 - 50x)\rho + (54 - 104x)\rho^2. \quad (24)$$

The last two coefficients are positive for $0 < x < 3/8$. Twice the constant term is

$$3 - 4x - 21x^2 + 52x^3.$$

For $0 < x \leq 1/4$, it is at least $3 - 4x - 21x^2 \geq 11/16$. For $1/4 \leq x < 3/8$, it is at least $3 - 4x - 8x^2 > 3/8$. Hence $N > 0$.

Equations (17), (21), and (22)–(24) show that $S_g < 5E/8$. Proposition 3.2 shows that this value is attained by an admissible selection on at most three labels. Together with the alternative preceding (14), this completes the proof. $\square$

6 Limitations and assistance disclosure

The theorem does not settle the five-label three-point assertion outside the $3 + 1 + 1$ direction stratum. In particular, systems with four or five distinct directions are untouched, as are other multiplicity patterns. It also gives no result for the six-label four-point assertion. Thus both of the broader finite-support questions, and Zhang’s seven-atom conjecture, remain unresolved.

OpenAI Codex assisted during proof development with case organization, algebraic cross-checks, numerical stress tests, and preparation of the manuscript. Numerical tests were used only as diagnostics and are not part of the proof. No formal proof assistant was used; all arguments needed for the stated theorem are displayed above.

References

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